

Probability and Statistics – for Computer Science

Based on Forsyth, *Probability and Statistics for Computer Science* (2018)

## Applied Statistics and Probability for Computer Scientists

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# The Significance of Evidence

Chapter 7 — Hypothesis Tests,  $t$ -Tests, and P-Value Hacking

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## Chapter Outline | What we will cover

The Logic of Hypothesis Testing  
Errors, Power, and Confidence  
Comparing Two Means:  $t$ -Tests  
Other Useful Tests

P-Value Hacking  
Summary

### Learning Objectives

- ▶ Explain the logic of **hypothesis testing** and  $p$ -values
- ▶ Choose and apply the right  $t$ -test for comparing means
- ▶ Use **F-tests** and  $\chi^2$  tests for variances and fit
- ▶ Recognize and avoid  $p$ -hacking



## Section 1

# The Logic of Hypothesis Testing

## The framework

- ▶  $H_0$  (**null**): "nothing interesting is happening"
- ▶  $H_1$  (**alternative**): the effect we suspect

## The logic

Assume  $H_0$  is true. If the observed data would be *very unlikely* under that assumption, reject  $H_0$  in favor of  $H_1$ .

## P-Values: The Core Concept | How surprising is the data, if $H_0$ is true?

### Definition

The  $p$ -value is the probability, *assuming*  $H_0$ , of seeing a result as extreme or more extreme than what was observed.

### Common misconception

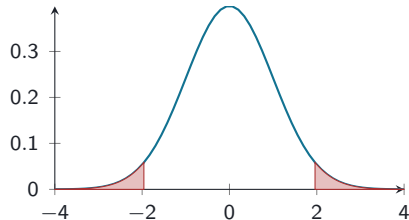
The  $p$ -value is **not** "the probability  $H_0$  is true," and  $1 - p$  is **not** "the probability  $H_1$  is true."

# Significance Levels | The $\alpha$ threshold and the decision rule

## Decision rule

Fix  $\alpha$  (often 0.05) *before* looking at the data:

$$\text{reject } H_0 \iff p < \alpha$$



Rejection region for a two-tailed test,  $\alpha=0.05$

## One-Tailed vs. Two-Tailed Tests | Which direction are you asking about?

| Type       | Question                                                             |
|------------|----------------------------------------------------------------------|
| Two-tailed | Is the parameter <i>different</i> from a value (either direction)?   |
| One-tailed | Is the parameter <i>greater than</i> (or <i>less than</i> ) a value? |

### Choose before, not after

The direction must be decided *before* seeing the data — choosing it afterward inflates your effective  $\alpha$ .

## Section 2

# Errors, Power, and Confidence

## Types of Errors | Two ways to be wrong

|                | $H_0$ true                | $H_0$ false               |
|----------------|---------------------------|---------------------------|
| Reject $H_0$   | Type I error ( $\alpha$ ) | Correct (power)           |
| Fail to reject | Correct                   | Type II error ( $\beta$ ) |

### Power

Power =  $1 - \beta$ : the probability of correctly detecting a real effect. Increases with larger  $n$ , larger effect size, or larger  $\alpha$ .

## Worked Example: Coin Fairness Test | Applying the full recipe

$H_0 : p = 0.5$ . Flip  $n = 100$  times, observe 62 heads.

$$z = \frac{0.62 - 0.5}{\sqrt{0.5(0.5)/100}} = 2.4$$

### Conclusion

Two-tailed  $p$ -value  $\approx 0.016 < 0.05$  — reject  $H_0$ : the coin looks biased.

### CIs and tests are two views of the same coin

Rejecting  $H_0 : \mu = \mu_0$  at level  $\alpha \iff$  the  $(1 - \alpha)$  confidence interval for  $\mu$  excludes  $\mu_0$ .

### Section 3

# Comparing Two Means: $t$ -Tests

# The Two-Sample Problem | Do two groups have different means?

## Hypotheses

$$H_0 : \mu_1 = \mu_2 \quad H_1 : \mu_1 \neq \mu_2$$

## Which test?

The right test depends on whether variances are equal, and whether the samples are paired.

## Pooled $t$ -Test (Equal Variances) | Combine both samples' spread

### Pooled variance

$$s_p^2 = \frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}$$

### Test statistic

$$t = \frac{\bar{X}_1 - \bar{X}_2}{s_p \sqrt{1/n_1 + 1/n_2}}$$

with  $n_1 + n_2 - 2$  degrees of freedom.

## Welch's $t$ -Test (Unequal Variances) | The safer default

### Test statistic

$$t = \frac{\bar{X}_1 - \bar{X}_2}{\sqrt{s_1^2/n_1 + s_2^2/n_2}}$$

### Degrees of freedom

Welch–Satterthwaite approximation — usually non-integer. Use this test *by default*: it costs little when variances are equal and protects you when they are not.

## Paired $t$ -Test | When observations come in natural pairs

### Key idea

Reduce to a *one-sample* test on the differences  $d_i = X_{1i} - X_{2i}$ .

### Test statistic

$$t = \frac{\bar{d}}{s_d / \sqrt{n}}, \quad n - 1 \text{ d.f.}$$

e.g. before/after measurements on the *same* subjects.

## Choosing the Right Test | A quick decision guide

| Situation                            | Test                             |
|--------------------------------------|----------------------------------|
| Same subjects, before/after          | Paired $t$ -test                 |
| Independent groups, similar spread   | Pooled $t$ -test                 |
| Independent groups, different spread | Welch's $t$ -test (safe default) |

## Effect Size: Cohen's $d$ | Statistical significance is not the whole story

### Definition

$$d = \frac{\bar{X}_1 - \bar{X}_2}{s_p}$$

### Interpretation

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$|d|$  Size

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0.2 Small

0.5 Medium

0.8 Large

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### Why it matters

A huge sample can make a *tiny, unimportant* difference "significant" — always report effect size alongside  $p$ .



# Section 4

## Other Useful Tests

## F-Tests for Variances | Are two groups equally variable?

### The $F$ -statistic

$$F = \frac{s_1^2}{s_2^2} \sim F_{n_1-1, n_2-1}$$

### Decision rule

Reject "equal variances" if  $F$  is far from 1 (too large or too small) relative to the  $F$ -distribution's critical values.

## $\chi^2$ Goodness-of-Fit Test | Does the data match a model?

### The statistic

$$\chi^2 = \sum_i \frac{(O_i - E_i)^2}{E_i}$$

### Also used for independence

The same statistic tests whether two categorical variables in a contingency table are independent.

### Requirement

Expected counts  $E_i$  should generally be  $\geq 5$  per cell for the  $\chi^2$  approximation to be valid.

## Section 5

# P-Value Hacking

# The Jelly Bean Problem | Why testing 20 things finds "significance" by chance

## The setup

Test 20 jelly bean colors for a link to acne, each at  $\alpha = 0.05$ . None actually matter.

## The math

$$P(\text{at least one hit}) = 1 - (1 - 0.05)^{20} \approx 0.64$$

## The multiple comparisons problem

The more tests you run, the more likely *some* will look significant by pure chance — correct with methods like Bonferroni.

| Form                        | What it looks like                                         |
|-----------------------------|------------------------------------------------------------|
| <b>Multiple comparisons</b> | Testing many outcomes, reporting only the "hit"            |
| <b>Optional stopping</b>    | Peeking at data, stopping as soon as $p < 0.05$            |
| <b>Researcher d.o.f.</b>    | Trying many analysis choices, keeping the best one         |
| <b>HARKing</b>              | Hypothesizing After Results are Known, dressed as a priori |

# The ASA Statement on P-Values | What a $p$ -value is not

## Not measures of

- ▶ The probability  $H_0$  is true
- ▶ The size or importance of an effect
- ▶ A sole basis for a scientific conclusion

## Best practices

Pre-register hypotheses, report effect sizes and CIs, correct for multiple testing, and value **replication**.

| Approach             | What it adds                                   |
|----------------------|------------------------------------------------|
| Confidence intervals | A range, not just a yes/no verdict             |
| Bayesian methods     | Direct probability statements about hypotheses |
| Effect sizes         | Practical, not just statistical, significance  |
| Replication          | The strongest evidence of all                  |



## Section 6

# Summary

# The Hypothesis Testing Recipe | Six steps, every time

1. State  $H_0$  and  $H_1$  *before* seeing the data
2. Choose  $\alpha$  in advance
3. Pick the appropriate test statistic
4. Compute the test statistic and its  $p$ -value
5. Compare  $p$  to  $\alpha$ ; decide
6. Report the effect size alongside the decision

## Chapter Summary | Key takeaways

### Test

$p$ -values quantify surprise under  $H_0$ ;  $t$ -,  $F$ -, and  $\chi^2$ -tests cover means, variances, and fit.

### Report

Always pair a  $p$ -value with an **effect size** and, ideally, a confidence interval.

### Avoid

$p$ -hacking: peeking, multiple untracked comparisons, and HARKing all inflate false discoveries.

### Looking ahead

Ch. 8: designing **experiments** with ANOVA to compare more than two groups at once.

# Thank you!

Questions & Discussion

Next: Chapter 8 — Experiments